he could just as easily have rejected any of his other assumptions about the world. As Quine said, our beliefs about the world can only be tested against the world as a group, not individually, and this holds for falsification just as much as for verification. No theory, in isolation, is falsifiable.

The history of science bears this out: when an experimental or observational result doesn't match a theoretical prediction, often one of the auxiliary assumptions used to generate the prediction is discarded, rather than the "main" theory itself. In 1781, John Herschel discovered Uranus, and astronomers of the time immediately set about predicting its motion using the cutting-edge science of Isaac Newton's gravitation and laws of motion. Over the course of the next few decades, as more observations poured in and calculations were refined, several astronomers realized that Uranus was not, in fact, moving as Newton's law of universal gravitation said it should be. But rather than throwing out Newtonian gravity as falsified by observation, they theorized that there was another as yet unseen planet, even farther out than Uranus, that was causing the anomalies in Uranus's motion. One of these astronomers, Urbain Le Verrier, calculated exactly where to find this planet, and, in 1846, a group of German astronomers found Neptune exactly where Le Verrier said it would be. So, rather than being falsified, Newtonian gravity lived to fight another day. And when, several years later, Le Verrier and others noticed that Mercury, the closest planet to the Sun, was not moving as it should, they once again did not throw out Newtonian gravity and instead postulated another new planet, so close in to the Sun that it was perpetually lost in its glare. They named this hypothetical scorched planet "Vulcan," after the Roman god of the forge, and immediately set about looking for it. They hunted for it during solar eclipses, when the Sun was masked by the Moon. Several teams, including one led by Le Verrier himself, claimed to have found the elusive Vulcan, but the planet was never conclusively pinned down. Finally, in 1915, Albert Einstein proved Vulcan was a phantom: his new theory of general relativity explained the motion of Mercury perfectly, without invoking a new planet. Newtonian gravity had been wrong all along—but it took a new theory to displace it, rather than an alleged "falsification."

Even Popper understood that falsification couldn't be the acid test for scientific theories: he admitted that no theory could be falsified in isolation but suggested that good scientists would reject their own theories rather